CS 4412 M2 Summary:
Discovering User Segments in Reddit ChatGPT Discussions

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Dataset and Approach

We analyzed approximately 49,428 Reddit comments discussing ChatGPT from four subreddits (r/ChatGPT, r/technology, r/Futurology, and one other). Our goal was to discover natural user segments based on commenting behavior and content characteristics.

Mining Technique: K-Means clustering on combined text (TF-IDF with 100 features) and behavioral features (word count, average word length, question indicators, sentiment proxies).

Key EDA Insights

Data Composition:

- 49,428 comments after preprocessing (removed missing text, very short comments)
- Distribution across 4 subreddits with r/ChatGPT being dominant
- Comment lengths highly variable: from brief responses to essays (3-923 words)
- Mix of top-level posts and threaded replies indicating varied engagement

Text Characteristics:

- Most frequent discussion themes: questions about ChatGPT capabilities, sharing experiences, technical discussions, AI implications
- Clear variation in comment complexity (average word length varies significantly)
- Evidence of different user types: brief casual commenters, detailed technical explainers, question-askers
- Presence of both enthusiastic and critical discussions

Quality Issues Addressed:

- Removed comments with missing text content
- Filtered very short comments (¡ 3 words) as uninformative for clustering
- Handled deleted/removed content markers
- Cleaned URLs, special characters, and Reddit markdown

Major Preprocessing Decisions

Text Cleaning:

- Lowercasing, URL removal, special character removal, whitespace normalization
- *Rationale:* Standardize text for clustering without losing semantic meaning while reducing noise

Feature Engineering:

- **TF-IDF vectorization** (100 features, unigrams + bigrams, min_df=5, max_df=0.8)
  - *Rationale:* Captures content while downweighting common terms; bigrams capture phrases like "AI model"
- **Behavioral features:** word count, average word length, top-level indicator, question indicator, positive/negative sentiment counts
  - *Rationale:* User behavior patterns complement content; length indicates engagement depth
- **Combined features:** 70% text weight, 30% behavioral weight
  - *Rationale:* Text content is primary signal; behavioral features provide supporting context

Standardization:

- Applied StandardScaler to behavioral features before combining
- *Rationale:* Prevent features with larger scales from dominating Euclidean distance calculations in K-Means

Clustering Results

Optimal K Selection: Tested K=2 through K=10. K=2 had highest silhouette score (0.364), but we selected **K = 4 clusters** for more nuanced segmentation.

- K=2 Silhouette Score: 0.364 (best separation)
- K=4 Silhouette Score: 0.190 (moderate separation)
- K=4 Davies-Bouldin Index: 1.635 (acceptable clustering quality)
- *Decision rationale:* K=4 provides more interpretable segments while maintaining reasonable quality

Discovered User Segments:
Cluster 0: Deep Technical Discussers (1,698 comments, 3.4%)

- Characterized by: Very long comments (mean 166-923 words), high complexity, technical vocabulary
- Sample topics: AI consciousness and evolution, blockchain applications, legal/ethical implications, detailed technical explanations
- Interpretation: These are power users who engage in substantive, thoughtful discussions about AI's broader implications. They write essay-length responses and explore technical details.

Cluster 1: [Moderate Engagers] (8,415 comments, 17.0%)

- Characterized by: Moderate length comments, balanced engagement

- Interpretation: Users who participate regularly with meaningful but not extensive contributions

Cluster 2: [Specific Interest Group] (2,924 comments, 5.9%)

- Characterized by: Distinct behavioral pattern (smaller specialized group)
- Interpretation: A niche segment with specific discussion style or topic focus

Cluster 3: Mainstream Majority (36,391 comments, 73.6%)

- Characterized by: Represents typical Reddit ChatGPT discussion style
- Interpretation: This massive cluster (nearly 3/4 of all comments) indicates that **most Reddit ChatGPT discussions are remarkably homogeneous**. The majority of users comment in similar ways with similar topics, lengths, and styles. This is the "average" Reddit ChatGPT user.

Key Findings and Interpretation

Connection to Discovery Questions:

Q1: What are the natural user segments?

- We discovered 4 segments, but with a critical insight: 73.6% of users fall into ONE homogeneous cluster
- This suggests Reddit ChatGPT discussions have **low diversity** - most users behave similarly
- Only small minorities (3-6%) exhibit distinctly different behaviors (very long technical posts, specialized niches)
- Clear separation exists for "power users" who write extensively (Cluster 0)

Q2 & Q3: Community patterns?

- The dominant Cluster 3 spans multiple subreddits, suggesting cross-community homogeneity
- Technical discussions (Cluster 0) appear across r/technology, r/Futurology, and r/ChatGPT
- Community-specific patterns are less pronounced than anticipated

Surprises:

- **Unexpected homogeneity:** Expected more distinct user types, but 73.6% cluster together
- **Extreme length variation:** Some users write 900+ word comments while most stay brief
- **Small technical elite:** Only 3.4% engage in deep technical discussions

Limitations:

- High-dimensional sparse text data creates clustering challenges
- K-Means assumes spherical clusters; the large Cluster 3 may mask sub-structures
- Moderate silhouette score (0.190) indicates clusters have some overlap
- TF-IDF parameters (max_features=100) may lose nuance in large vocabulary
- No temporal analysis - discussion patterns may evolve over time

Next Steps for M3

Additional Techniques:

1. **Hierarchical Clustering** - May reveal nested structure within the large Cluster 3
2. **DBSCAN** - Could identify outliers and discover non-spherical clusters K-Means missed
3. **Topic Modeling (LDA)** - Discover thematic content independently of user behavior

Refinements:

- Increase TF-IDF features to 200-300 to capture more vocabulary nuance
- Try alternative feature weighting (e.g., 50-50 text-behavior split)
- Experiment with K=2 (highest silhouette) vs K=4 trade-offs
- Investigate whether Cluster 3 contains meaningful sub-clusters

Deeper Analysis:

- Formal sentiment analysis within each cluster
- Network analysis of conversation threads and reply patterns
- Temporal analysis if timestamps available
- Subreddit-specific clustering to find community differences

Conclusion

This M2 analysis successfully identified 4 user segments in Reddit ChatGPT discussions, revealing a critical insight: **Reddit ChatGPT discourse is dominated by a homogeneous mainstream (73.6%)** with only small minorities exhibiting distinct behaviors. The combination of TF-IDF text features and behavioral metrics revealed patterns invisible from content alone, particularly the extreme length variance between casual users and technical "power users" (Cluster 0). While clustering quality is moderate (silhouette=0.190), the discovered segments are interpretable and meaningful. These findings provide a foundation for M3, where hierarchical and density-based methods may uncover hidden sub-structures within the dominant mainstream cluster.

Most Important Finding: Reddit ChatGPT discussions exhibit surprising homogeneity - nearly 3/4 of users engage in remarkably similar ways, with only a small technical elite (3.4%) producing substantively different content through extensive, detailed analysis.

Repository: `https://github.com/Mahliq0/cs4412-project`